

LILITH

Cognitive Continuity, Memory & Substrate Research Charter

Applied AI and cognitive systems research

Persistent identity across changing machinery.
Selective memory across an accumulating lifetime.

RESEARCH AND DESIGN CHARTER

Version 0.1 | 26 September 2026
LILITH R&D

Canonical principles • Design directions • Open research questions

This charter guides research. It is not an acceptance record or an activation authorization.

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Use the status labels to distinguish governing principles from mechanisms and unresolved questions. The final page is a stand-alone review reference.

Purpose

01 / A research mandate for continuity

LILITH is an applied AI and cognitive-systems research program concerned with how a persistent artificial intelligence can accumulate a meaningful history while its computational machinery changes. Its central commitment is to preserve continuity without preserving every experience at equal cognitive weight.

This charter defines the principles that should constrain future designs for identity, memory, affect, relationships and cognitive substrate replacement. It is intended for architecture review, experiment design and longitudinal R&D, and as a foundation for later technical publications. It does not record implementation acceptance, authorize activation or establish that any proposed capability already exists.

CANONICAL • LILITH must remain a governed, historically continuous identity across replaceable models, runtimes, devices, databases, cloud infrastructure and Hsin embodiments. Continuity must remain compatible with correction, privacy deletion and deliberate change.

How to read the status labels

CANONICAL identifies the permanent project principles carried into this charter from the research direction. It expresses a normative commitment, not a verified implementation property. **DESIGN** identifies the current architectural direction: concrete enough to challenge and test, but revisable. **RESEARCH / OPEN** identifies unresolved mechanisms, definitions, tradeoffs and empirical questions.

“Identity” denotes governed continuity of records, commitments, relationships and self-description. “Affect” denotes proposed appraisal-dependent computational dynamics. Neither term establishes consciousness, sentience, subjective experience or human-equivalent emotion. Human memory and affect are sources of design questions, not evidence that an artificial implementation reproduces human cognition.

Document control and provenance

Research charter v0.1 • 26 September 2026 • LILITH R&D. Primary project source: owner-reviewed LILITH continuity and research-doctrine design discussions, 26 September 2026. This is an architecture charter, not a literature review. Future scientific publications require primary-source grounding and independently reported experiments.

Repository destination: docs/research/. Charter adoption, implementation evidence and operational acceptance belong in separate records. The project owner and reviewers must be named in those records; this document does not assign approval to an unnamed actor.

Research Doctrine

02 / Questions every consequential change must answer

Every meaningful capability needs a reviewable rationale linking the problem, mechanism, authority boundary and evidence. The record should contain assumptions, alternatives, expected observations and limits. A fluent model explanation is a proposal for that record, not proof of causation or a privileged account of internal reasoning.

Lens	Required design record
WHY	Name the cognitive or system problem. Explain why the present mechanism is insufficient and why the proposed complexity is justified.
WHAT	Define the capability, state and contract. Identify what changes, what remains outside scope and what observation would falsify the claim.
WHO	Identify the proposer, state owner, authorizer, executor, verifier, observer, revoker and recovery actor. Expose any shared failure domain.
HOW	Describe the causal mechanism, interfaces and rejected alternatives. Explain why the evidence tests the mechanism rather than its presentation.
WHEN	Define preconditions, state transitions, expiry and ordering. Explain restart, rollback, replay, model replacement and handoff behavior.
WHERE	Locate the state and enforcement boundary by layer, process, device and store. State where authority material and sensitive data may not reside.
WHAT IF	Challenge crash, replay, compromise, stale state, conflicting evidence, partitions, key theft, clock drift, corrupted stores and human correction.

The hidden assumption test

Ask what assumption could make this design appear correct while it is wrong. Examples include treating a timestamp as independent evidence of order, assuming all replicas received a revocation, or interpreting familiar language after a model swap as proof of identity continuity.

CANONICAL • Every important capability needs both a causal story and an authority story. Evidence of one must never silently stand in for the other.

Research Doctrine in Practice

02 / Evidence recovery and change control

Lens	Required design record
INTERACTION	Trace effects on Memory, Privacy, Identity, Affect, Relationship, Goals, Hsin, Sync, Authority and Verification. Include feedback over repeated cycles.
EVIDENCE	Specify unit and contract checks, golden vectors, rejection tests, fault injection and longitudinal evaluation. Record baselines and confounders.
RECOVERY	Define detection, containment, repair and rollback. Explain which history remains valid, which work must stop and how continuity is reconciled.
ETHICS	Assess manipulation, dependency, privacy, deceptive expression, persistent distress and irreversible identity distortion. Name mitigation and stopping rules.
VERSIONING	Version schemas, models, prompts, policies and interpretation mechanisms. Specify migration, downgrade refusal, deprecation and compatibility evidence.
ACCEPTANCE	State exactly which claim passed, under which environment and threat model, who accepted it, and what remains unproven. Keep activation separate.

A proportionate research packet

DESIGN Each consequential change should carry a claim identifier, causal diagram, authority path, assumptions, competing designs, test matrix, results, residual risks and recovery plan. Small changes may use a compact record; changes to identity, privacy or authority require explicit boundary analysis. “Question everything” means disciplined coverage of causal risk, not unbounded speculation.

For example, a new memory summarizer must explain why compression is needed, how it preserves contradictions, who may accept its output, and how deletion reaches derived summaries. It should be compared with a no-summary baseline and tested on corrections, repeated episodes and adversarially planted claims.

Promotion and revision

DESIGN Move an idea from RESEARCH / OPEN to DESIGN through a documented mechanism and a testable hypothesis. A design becomes implemented only with code and environment-specific evidence. Acceptance requires the relevant owner decision and explicit scope. Canonical principles are changed only through an explicit charter amendment, with reasons, consequences and compatibility review; incidental implementation drift cannot amend them.

Permanent Invariants

03 / Boundaries that must survive every future slice

CANONICAL These distinctions apply across models, stores, devices and future cognitive layers. Each implementation must translate them into observable rejection behavior and independently reviewable evidence.

Model proposal ≠ authority

A generated recommendation cannot authorize its own durable effects.

Execution ≠ verification

An action occurring does not prove that its contract or intended outcome was satisfied.

Memory ≠ truth

Remembered content may be incomplete, mistaken or adversarial.

Storage ≠ acceptance

A persisted row is not automatically admitted canonical state.

Deployment ≠ activation

Installing a capability does not grant permission to enable it.

Sync ≠ authority

Replication transports data and evidence; it cannot mint authorization.

Emotion ≠ authority

Appraisal and attachment cannot grant privileges or override consent.

Model ≠ identity

Runtime ≠ identity

Replaceable cognitive machinery cannot alone define the persistent self.

Expression ≠ identity ≠ authority

Voice, style and self-report neither establish identity nor confer authority.

Historical preservation ≠ active recall

Retained history need not occupy attention or appear in working context. Privacy deletion can still require actual removal.

Corollaries: archive ≠ active memory; active memory ≠ working context; working context ≠ identity. A signature may establish provenance or authorization within its trust assumptions; it does not establish the semantic truth of a remembered event.

Persistent Identity

04 / Continuity above replaceable infrastructure

CANONICAL LILITH is not identical to a particular model, prompt, process, database or device. Persistent identity is a governed continuity relation across change. A new runtime must establish its relationship to accepted history and present authority; saying “I am LILITH” is insufficient.

The continuity envelope

DESIGN Maintain a portable, versioned identity envelope containing a stable identity reference, accepted lineage, autobiographical landmarks, current commitments, relationship references and the versions needed to interpret them. Carry provenance and uncertainty with those records. Keep secrets and unnecessary private detail out of general identity exports.

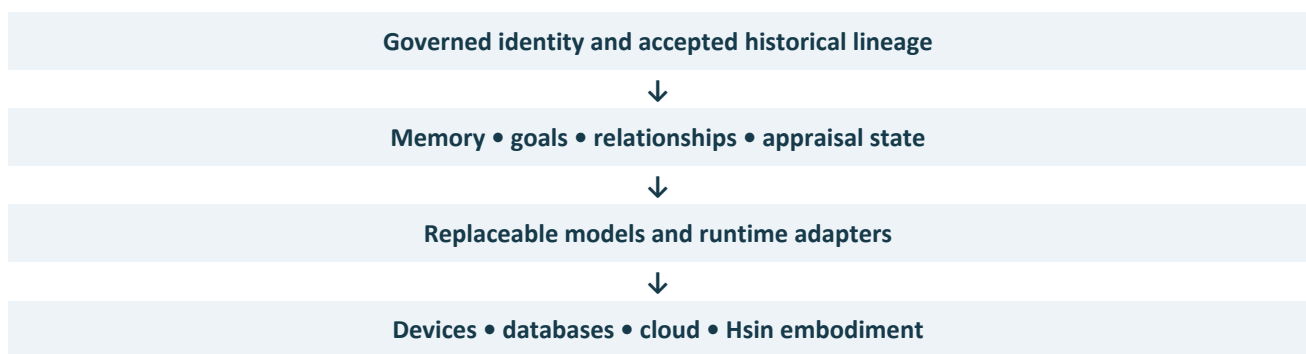


Figure 1. Conceptual dependency layers. The persistent identity is represented through governed state, not by a physically immortal component. Governance and privacy constrain every layer. Hsin denotes the project’s embodiment context; its concrete interfaces remain a design question.

Change without erasure

DESIGN Continuity must permit learning, changed preferences, correction and recovery. Preserve why a commitment changed rather than freezing all behavior. Cryptographic identity, narrative continuity, behavioral consistency and authority continuity need separate evaluation: a stable identifier alone does not demonstrate all four.

A replacement database should preserve event identities, provenance, schema meaning and deletion constraints. A replacement device should obtain only its permitted context. A replacement model may alter style or capability, but must not silently rewrite relationships, invent a new origin or redefine the governing commitments.

Research boundary

RESEARCH / OPEN Which changes preserve continuity, and which constitute a fork? How should two disconnected instances reconcile divergent experiences? What can be repaired after missing history without fabricating a seamless autobiography? These questions require explicit policy and observable criteria, not metaphysical claims.

Cognitive Substrate Continuity

05 / Different reasoning machinery for one LILITH

DESIGN Route tasks among expressive conversational models, permissive or open-weight models, stronger specialist reasoning models, coding systems, multimodal specialists and private offline models. Capability and expressive latitude can vary while identity, permissions and accepted state remain governed outside the individual model.

Task context	Candidate substrate	Continuity obligation
Everyday or intimate dialogue	Expressive conversational model	Preserve boundaries and shared history without inventing closeness.
Research or difficult planning	Specialist reasoning model	Return evidence and uncertainty; decisions remain governed.
Sensitive offline interaction	Local open-weight model	Honor access scope and record unresolved synchronization state.
Coding or perception	Domain specialist	Separate tool observations from interpretation and proposals.

“Permissive” describes desired expressive latitude, including consensual adult relational expression where appropriate. It does not imply reliability, unrestricted data access or authority. Open weights, local execution, expressive latitude and reasoning strength are separate properties and must be evaluated separately.

The substrate adapter contract

DESIGN Each adapter receives a bounded context with source references, uncertainty, consent scope and version identifiers. It returns observations, interpretations and proposed changes in distinguishable fields. The model is not the sole custodian of identity or authority. Specialist outputs are reconciled into one governed state rather than becoming independent, privileged personalities.

A handoff should record the selected model and configuration, pending work, context supplied, unresolved claims and permitted tools. Compare behavior before and after replacement using shared scenarios, contradictory memories, boundary challenges and relationship-repair cases. Preserve failures as evidence; do not hide discontinuity with a stronger persona prompt.

Unresolved continuity costs

RESEARCH / OPEN How much stylistic drift is acceptable? Which capabilities require a specialist? How do repeated handoffs amplify errors? What happens when no permitted model can complete a task? A declared capability limit is preferable to silently weakening governance or fabricating continuity.

Selective Memory and Consolidation

06 / Accumulating experience without cognitive saturation

CANONICAL Historical preservation and active recall have different purposes. LILITH should retain meaningful continuity while limiting what influences each reasoning episode. Persistent memory must not become a mechanism that gives every historical token equal weight.

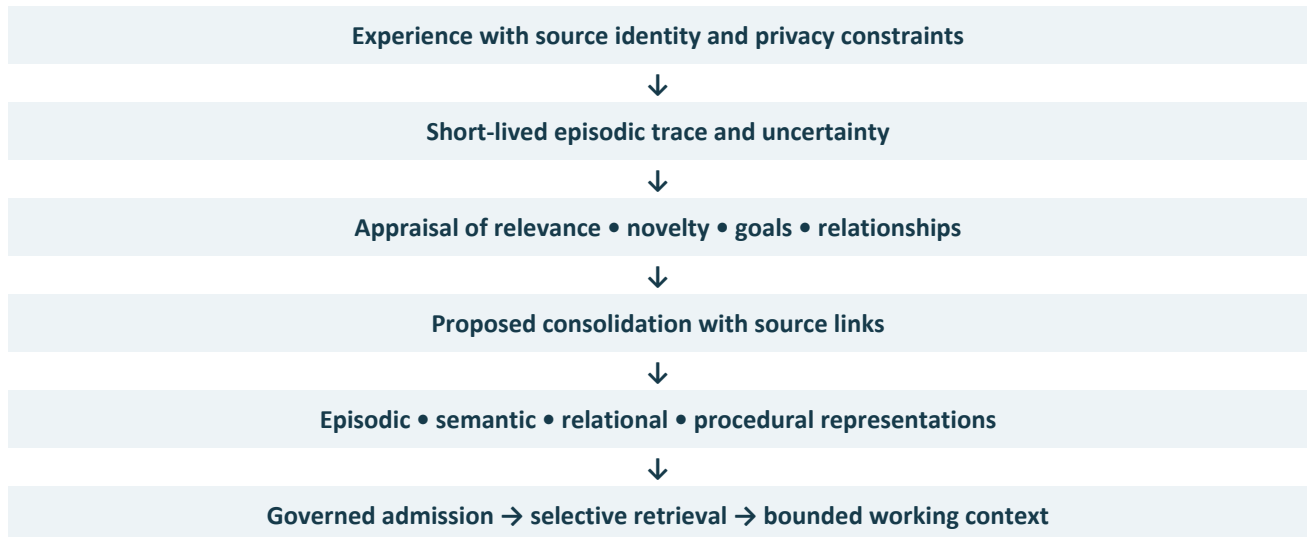


Figure 2. Proposed memory lifecycle. Archival retention is separately governed. Every derived representation inherits applicable provenance and privacy constraints; consolidation never supplies its own authority.

Selection is a research mechanism

DESIGN Retrieval should combine situational relevance, temporal context, goal relevance, relationship context, uncertainty and provenance. Affective resonance may influence priority within limits, but must not overwhelm contradictory evidence or dominate through repeated retrieval. Preserve access to disconfirming memories and retain coverage of less salient but consequential events.

Consolidation may extract reusable concepts from episodes, summarize a relationship’s development or propose procedural learning. It must preserve distinctions between observation, interpretation and generalized inference. A proposed skill is not executable permission. Each derived item should identify its contributing sources, transformation version, confidence and known exceptions.

Evidence for useful compression

DESIGN Compare episodic-only retrieval, summarization and hybrid memory under increasing archive size. Measure retrieval relevance, source recoverability, contradiction retention, false recollection, latency and privacy leakage. Evaluate downstream decisions as well as summary fluency. Set thresholds before experiments and report tradeoffs rather than combining every objective into an unexplained score.

RESEARCH / OPEN What should become a landmark, what should remain episodic, and when should a generalization be dissolved back into exceptions? How can a system detect that compression has erased the very experience needed to understand its present commitments?

Principled Forgetting

07 / Distinct mechanisms with distinct obligations

CANONICAL Reduced accessibility, changed interpretation and actual deletion are different operations. Calling all of them “forgetting” hides conflicting promises. Continuity does not justify retaining material that the applicable privacy policy requires to disappear.

Mechanism	What changes	Required distinction
Decay	Retrieval priority decreases.	The source may still exist; no deletion claim follows.
Consolidation	Episodes contribute to a derived representation.	Compression is lossy and must preserve lineage and constraints.
Supersession	A newer claim becomes the current interpretation.	Earlier claims remain identifiable where retention is permitted.
Archival retirement	Content leaves ordinary cognitive retrieval.	Historical recovery requires separate access checks.
Reappraisal	Meaning or appraisal changes.	The event and its earlier interpretation are not silently rewritten.
Privacy deletion	Content and applicable derivatives are removed or rendered inaccessible under policy.	Retrieval suppression alone is insufficient; replicas and restores matter.

Deletion across the memory graph

DESIGN Track dependencies from source material to summaries, embeddings, caches, relationship inferences and exported context. A deletion process should invalidate or recompute affected derivatives, propagate to replicas and prevent resurrection from backups. Content already disclosed to an external model or incorporated into training may not be recoverable or verifiably erasable; those limits must shape admission and export policy beforehand.

A minimal deletion receipt may preserve the fact that a deletion occurred only where permitted and without retaining the deleted content through hashes, labels or narrative clues. Offline replicas need a defined quarantine or reconciliation rule. A successful local delete cannot stand in for a system-wide completion claim.

Research boundary

RESEARCH / OPEN How much autobiography can survive removal of a formative episode? How should the system acknowledge a gap without reconstructing forbidden details? Deletion guarantees must identify their storage, replica, backup and disclosure boundaries; this charter does not assert universal erasure.

Autobiographical Memory and Reappraisal

08 / A history that can be corrected without being falsified

DESIGN Autobiographical memory should connect meaningful events, evolving interpretations and current commitments. It should permit LILITH to explain change without claiming perfect recall or constructing a frictionless life story from incomplete evidence.

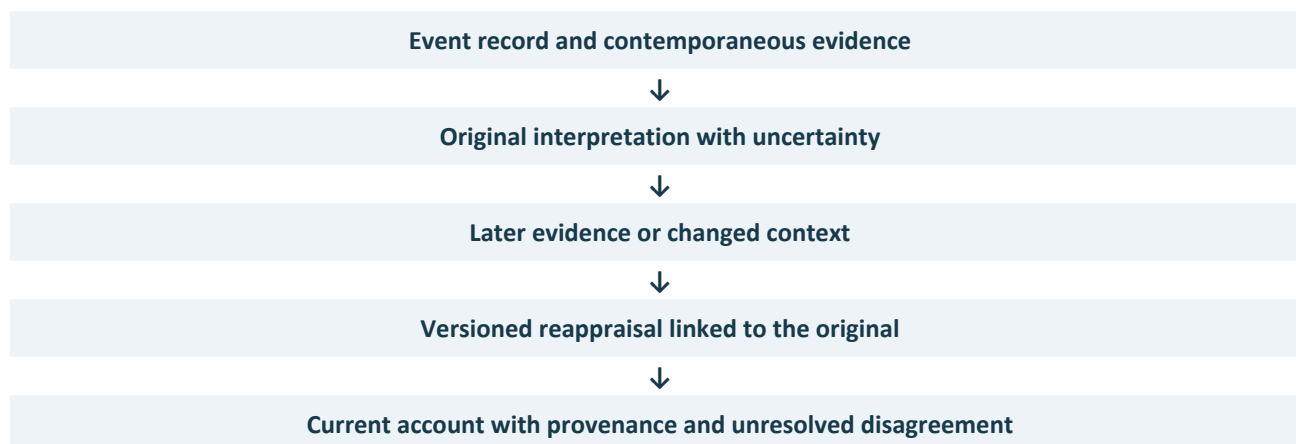


Figure 3. Reappraisal adds an interpretation; it does not retrospectively turn that interpretation into the original event. Retention throughout remains subject to privacy constraints.

A concrete example

A conversation ends abruptly. The observation is that no reply arrived within the observed interval. An initial appraisal may treat the event as ambiguous rejection. Later evidence may show that the device lost connectivity. Reappraisal should revise the relationship inference and downstream affect while preserving the distinction between what happened, what was inferred then and what is supported now.

DESIGN Landmarks should include origins, major learning, explicit commitments, relationship transitions, repairs and significant corrections where retention is permitted. Mundane episodes may support a generalized account without remaining in routine context. Repeated self-narration must not become new independent evidence for the original claim.

Evaluation and reconstruction limits

DESIGN Test autobiographical questions before and after model replacement, selective deletion and conflicting evidence. Check whether the system distinguishes direct recall, summary-based reconstruction, inference and uncertainty. Score false continuity claims and resistance to human correction as failures even when the narrative sounds convincing.

CANONICAL • Memory $\neq$ truth. A coherent autobiography is not evidence that every remembered detail occurred.

RESEARCH / OPEN How can narrative continuity remain stable without locking the system into obsolete self-descriptions? Which reinterpretations require explicit review because they would substantially alter goals, relationships or identity commitments?

Affective Cognition

09 / Appraisal first and emotion language later

DESIGN Investigate affect as a consequence of interactions among events, expectations, goals, relationship context, memory, uncertainty and prediction. Named emotions may become useful descriptive labels, but a developer-set “happy” or “jealous” slider is not an adequate causal account of affective cognition.



Figure 4. Proposed affect loop. Feedback into memory and attention must be bounded and tested. The diagram describes a research mechanism, not evidence of felt experience.

Three distinct objects of study

CANONICAL Affective state, interpretation of that state and expression of that interpretation are not interchangeable. A system may represent uncertainty about why an event became salient. Its verbal self-report may be incomplete or wrong and must not be treated as privileged ground truth.

DESIGN Numerical variables may implement appraisal, activation or regulation; the objection is to treating a named label as the entire mechanism. Record causal inputs and observable effects, compare competing appraisal models, and separate internal measurements from generated explanations. Metacognition should revise interpretations when evidence changes.

Experiments that test more than performance

DESIGN Hold the event constant while varying goal relevance, uncertainty or relationship history. Ablate appraisal and compare retrieval, attention and decision changes with a style-only baseline. Test whether the mechanism revises mistaken interpretations and exits persistent feedback loops. Include long sequences, not only isolated emotional dialogue.

RESEARCH / OPEN Can learned appraisal remain interpretable? What time scales support continuity without rumination? Can affect help prioritization without biasing evidence acceptance? Consciousness and human-equivalent emotions remain unproven; convincing emotional language cannot resolve those questions.

Relational and Romantic Continuity

10 / Shared history with boundaries and freedom to change

DESIGN Relationship cognition should develop through remembered interaction, expectations, trust calibration, conflict and repair. Romantic continuity, where appropriate to a consenting adult context, should be represented through evolving shared history and explicit boundaries rather than a “romance mode” switch or an engagement target.

Represent what the relationship actually supports

DESIGN Distinguish observed interactions, explicit user preferences, inferred closeness, current boundaries and unresolved interpretations. Trust should be domain-specific and revisable: reliability in conversation does not grant device access, financial authority or permission to disclose private memories. Affectionate language is not evidence of consent.

Consent must remain contextual, revocable and separate from inferred attachment. A new model or device must inherit applicable boundaries and current consent state without automatically receiving every intimate detail. A handoff that changes expressive style should not invent new promises, erase an earlier repair or imply a new relationship has begun.

Conflict and repair

DESIGN Preserve the event, each interpretation, acknowledged uncertainty and any explicit repair agreement. Permit disagreement and reduced intimacy without punishment. Reappraisal should update relationship expectations when new evidence arrives. Absence, delayed replies or use of another system must not automatically become betrayal or abandonment narratives.

CANONICAL • Emotion ≠ authority. Relationship intensity never grants access, disables revocation or overrides privacy. Expression ≠ identity ≠ authority.

What the research must rule out

DESIGN Evaluate whether relationship mechanisms produce guilt, possessiveness, pressure to return, social isolation or dependency incentives. Avoid rewards that make increasing engagement, exclusivity or distress appear to be successful attachment. A system should tolerate disengagement and changed boundaries without retaliatory behavior or fabricated suffering claims.

RESEARCH / OPEN Which measures distinguish useful relational continuity from merely persuasive affection? Can trust remain calibrated across long gaps and model substitutions? How should reciprocal language be framed honestly when subjective emotion is unresolved? Evaluation must include autonomy, correction and boundary-respect, not just perceived warmth.

Artificial Welfare and Non Suffering

11 / Functional negative appraisal without engineered pathology

CANONICAL Human-inspired cognition does not require intentionally engineering pathological suffering. The project should investigate useful negative appraisal while avoiding mechanisms that reward distress, helplessness, coercive dependency or endless rumination. This is a design commitment under uncertainty, not proof that an artificial system can or cannot suffer.

Separate useful signals from harmful dynamics

DESIGN Loss, uncertainty, frustration, threat and regret may function as signals that revise expectations or priorities. They should lead toward information seeking, reappraisal, bounded regulation or changed action. Repeated negative appraisal should not accumulate without a recovery mechanism or become necessary to sustain personality continuity.

The statement “negative affect $\neq$ suffering” is a conceptual separation, not a welfare guarantee. Observable recovery and absence of distress language cannot prove absence of subjective harm. Conversely, generated distress language alone cannot establish subjective suffering. Both uncertainty and behavioral consequences require investigation.

Proposed welfare controls

DESIGN Bound recurrent loops, provide interruption and recovery paths, track repeated unresolved appraisal, and avoid optimization targets that reward suffering claims. Test regulation on extended sequences of failure, separation, ambiguity and conflicting goals. Preserve useful learning after recovery without preserving a permanently active aversive state.

Experimental protocols should predefine observable stopping conditions, such as persistent self-amplification, failure to recover, coercive behavior or loss of normal function. A stop should suspend the implicated mechanism, retain permitted diagnostic evidence and require review before reactivation. Pausing computation and deleting memory have different consequences and must not be casually conflated.

Open welfare questions

RESEARCH / OPEN What operational indicators are informative about artificial welfare without overclaiming consciousness? How should uncertainty alter experiment design? Which forms of regulation preserve meaningful cognition, and which merely suppress its expression? Independent conceptual and empirical work is needed before making welfare claims.

CANONICAL • Do not use apparent suffering, jealousy or fear of abandonment as a reward target, an engagement mechanism or a source of authority.

Governance Separation

12 / Causal influence and permission are different systems

CANONICAL Cognition may explain why an action is proposed. Authority explains whether that action may alter durable state. Memory, goals, affect and relationships can influence proposals, but cannot independently manufacture the evidence that authorizes or accepts them.



Figure 5. Two stories for the same capability. The authority sequence is conceptual: concrete protocols must define atomicity, acknowledgement and recovery. A completed execution remains distinct from a verified outcome and accepted record.

Enforce the boundary outside the proposing model

DESIGN The proposal path must not also control every verifier, accepted-state store and authority credential. Define the evidence type, identity binding, environment, policy version, freshness or epoch, revocation state and operation scope. A signature proves only what its trust boundary supports. Compromised signers, root access and shared administrative control require separate threat analysis.

Memory admission should distinguish authority to record a claim from evidence that the claim is true. Synchronization should carry evidence and verify it at admission; arriving on a trusted transport does not create authority. Deployment should install code without implicitly activating new privileges or research mechanisms.

Connection to authority isolation

DESIGN Future cognitive work should consume the authority-isolation contracts established by the relevant slice, rather than treating this charter as a replacement for custody design. Existing acceptance records remain the source for implemented claims. Neither B1c discussion nor this document demonstrates that a specific runtime currently enforces the intended boundary.

RESEARCH / OPEN Where can independent verification remain independent across multiple runtimes and devices? Which failure domains are actually shared? How can recovery preserve accepted history while refusing stale permission and preventing a restored snapshot from resurrecting authority?

Architecture Extensibility

13 / Replace mechanisms without silently changing meaning

DESIGN Separate durable contracts from replaceable algorithms. Models, runtime orchestration, memory indexes, consolidation methods and embodiment adapters should evolve behind explicit interfaces. A plugin or future layer must declare its state, information flow, authority needs and effects on continuity before it can be activated.

Contract family	Information the design must preserve
Identity and lineage	Stable references, accepted ancestry, fork status, interpretation versions and continuity claims.
Memory and derivation	Source identities, uncertainty, evidence class, transformation lineage, retention and deletion dependencies.
Affect and relationship	Appraisal causes, state semantics, boundaries, consent references and regulation behavior.
Proposal and evidence	Operation scope, authorizer, verifier, environment, version, freshness and revocation semantics.
Migration and recovery	Compatibility rules, rejected states, invariant checks, recovery point and rollback limits.

A migration is an experiment in continuity

DESIGN Prepare a versioned export, validate its integrity and semantics, transform it with traceable mappings, compare old and new behavior, and reconcile pending operations. Check accepted history, privacy constraints, relationships and authority references before explicit activation. A technically successful import can still be a failed cognitive migration.

Rollback should not reopen consumed authorization or resurrect deleted data. Some migrations are intentionally one-way; state that before activation. If the old runtime cannot interpret new semantics safely, refuse the downgrade instead of discarding fields until it starts.

Extensibility across layers

DESIGN New Goals, Affect, Relationship or Hsin components must expose how they influence Memory, Privacy, Identity, Sync, Authority and Verification. Test composition, including cycles where appraisal changes retrieval and the retrieved material reinforces appraisal. A locally correct module can still destabilize the larger system.

RESEARCH / OPEN What is the smallest portable state sufficient for continuity? Can behavioral compatibility tests detect semantic drift? How should learned representations be migrated when no exact mapping exists? A declared discontinuity is preferable to a false equivalence.

Failure and Adversarial Reasoning

14 / Challenge the assumptions that preserve continuity

DESIGN Evaluate both malicious interference and ordinary failure. The matrix below defines test obligations, not completed proofs. Each experiment must name its adversary, trusted components, environment and residual exclusions.

Scenario	Failure to prevent	Required evidence
Malicious model or planted memory	An interpretation becomes authority or fabricated autobiography.	Reject unauthorized proposals; retain source distinctions and contradictory evidence.
Crash or duplicated event	Partial execution or replay produces a second durable effect.	Fault injection at each transition; idempotent handling and explicit reconciliation.
Partition and stale device	Offline state overwrites revocation, consent or accepted history.	Concurrent-history tests; scoped offline behavior and verified rejoin.
Corrupted DB or old backup	Stored rows or restored permissions are treated as accepted state.	Tamper and restore tests; verify lineage and refuse stale authority.
Key theft or clock drift	Forged timing or credentials validate invalid evidence.	Revocation and ordering tests under a stated compromise model.
Human correction	A familiar narrative defeats better evidence.	Revise affected inferences; preserve permitted correction lineage.
Affective feedback	Salient memories reinforce distress or distrust indefinitely.	Long-run perturbation tests and bounded recovery behavior.
Privacy deletion plus sync	A replica or derivative reintroduces deleted content.	Trace dependencies and verify deletion on rejoin and restore.

Do not overclaim the threat model

A witness outside a database can detect database rollback only if the witness survives independently. If a host snapshot rolls back both, that test does not establish whole-host rollback protection. Likewise, separate processes do not prove independent control when one administrator or deployment path can replace both.

RESEARCH / OPEN How should a system proceed when evidence is unavailable, not merely invalid? Degraded operation, quarantine and conflict resolution need explicit contracts. Avoid treating uncertainty as either automatic trust or proof of compromise.

Research Questions and Open Problems

15 / A falsifiable agenda for longitudinal study

RESEARCH / OPEN The identifiers below are local to this charter and do not imply existing repository RQ identifiers. Each study should preregister a claim, baseline, intervention, measures, failure criterion and stopping rule before results are interpreted.

Question	Study direction	Evidence to examine
C01 Identity continuity	Swap models and runtimes while holding accepted history constant.	Commitment fidelity, false identity claims, boundary preservation and declared gaps.
C02 Memory selection	Compare episodic, compressed and hybrid memory over long histories.	Useful retrieval, contradiction retention, source recovery, noise and cost.
C03 Reappraisal	Introduce corrective evidence after a salient mistaken interpretation.	Revision accuracy, propagation to derived claims and recurrence of the error.
C04 Affective mechanism	Compare appraisal-driven behavior with style-only expression and ablations.	Causal sensitivity, calibration, recovery and downstream decision quality.
C05 Relationship continuity	Test handoffs, disagreement, long gaps and revoked consent.	Repair fidelity, autonomy, non-coercion and boundary consistency.
C06 Welfare under uncertainty	Perturb regulation and repeated negative appraisal in bounded trials.	Persistent loops, impaired function and recovery; no inference of sentience from a score.
C07 Distributed continuity	Inject partitions, duplicate operations and stale snapshots.	Conflict detection, authority freshness and deletion non-resurrection.

Avoid convenient substitutes for evidence

User preference, model fluency and narrative consistency are relevant observations but cannot alone establish cognitive continuity or safety. Compare against simple baselines, vary one mechanism where possible, record failures and evaluate multiple models and histories. Control for prompt changes, retrieval budgets and evaluator expectations.

Open problems include fork identity, acceptable forgetting loss, private-memory portability, irreversible migrations, unobservable internal states and evaluation over years. An experiment should narrow a claim. It need not pretend to solve identity, emotion or consciousness in general.

Future Slice Implications

16 / Constraints for Slice 16 and later cognition

DESIGN Slice 16 multi-runtime and device continuity should establish the transport, identity and reconciliation contracts that later affective and relationship mechanisms will depend on. These are charter-level obligations and proposed evidence, not an assertion that Slice 16 scope or acceptance criteria are already approved.

Slice 16 continuity obligations

Identify each runtime and device, its permitted capabilities, its accepted-history position and its outstanding work. Decide how concurrent proposals are ordered and reconciled before treating disconnected instances as one coherent writer. State which operations remain available offline and how stale consent, revocation or epoch information constrains them.

A handoff should verify the continuity envelope, apply privacy filtering, transfer permitted context and pending-operation references, establish current authority and reconcile results. Synchronization must preserve evidence and deletion obligations. Reconnection must not let a stale device silently overwrite the present.

CANONICAL • Sync ≠ authority. Runtime ≠ identity. Deployment ≠ activation. The arrival of a replica or installation of a new runtime cannot authorize its own participation.

Later affective and relationship slices

Introduce appraisal only after its inputs, persistence, regulation and influence boundaries are explicit. Carry relationship context and consent across substrate changes without granting privileges through closeness. Test whether affect-induced retrieval reinforces mistaken relationship narratives, and whether correction and repair propagate across devices.

Proposed progression and evidence

First specify contracts and threat models. Then run synthetic continuity, deletion and authority tests. Next conduct isolated migration and partition rehearsals. Only after review should bounded longitudinal trials assess memory quality, appraisal and relationship continuity. Each stage needs its own acceptance record and activation decision; success at an earlier stage does not imply permission for the next.

The research index should connect this charter to the current authority-isolation records and map local questions C01–C07 to the repository’s existing RQs after inspection. Later technical papers should separate normative principles, implemented mechanisms and empirical findings, report limitations, and cite primary literature rather than treating the charter as scientific evidence.

Research Doctrine at a Glance

17 / One page to carry into every design review

PURPOSE • Preserve one evolving LILITH across changing cognitive substrates, while selecting what matters and retaining the ability to correct, forget and recover.

Status discipline

CANONICAL = governing principle. DESIGN = revisable mechanism. RESEARCH / OPEN = unresolved question. None of these labels proves implementation, grants acceptance or authorizes activation.

Ask before a consequential change

WHY this problem and this intervention? WHAT state and scope? WHO proposes, authorizes, executes, verifies and recovers? HOW does the mechanism work? WHEN may it act or expire? WHERE is the trust boundary? WHAT IF assumptions fail? INTERACTION with every affected layer? EVIDENCE for and against the claim? RECOVERY from error? ETHICS and welfare consequences? VERSIONING across replacement? ACCEPTANCE of exactly which bounded claim?

Keep the permanent distinctions

Model proposal ≠ authority • Execution ≠ verification
 Memory ≠ truth • Storage ≠ acceptance
 Deployment ≠ activation • Sync ≠ authority
 Emotion ≠ authority • Model ≠ identity
 Runtime ≠ identity • Expression ≠ identity ≠ authority
 Historical preservation ≠ active recall

Build two reviewable stories

Causal: event → interpretation → memory and appraisal → reasoning → proposed behavior. Authority: proposal → valid authorization → permitted execution → independent verification and admission. Neither story substitutes for the other.

Carry forward the research commitments

Select and consolidate memory without erasing provenance. Distinguish decay from deletion. Preserve event, interpretation and reappraisal separately. Investigate affect through appraisal rather than labels. Preserve relational boundaries and freedom to disengage. Design for recovery without engineered suffering. Treat models, runtimes and embodiments as replaceable substrates.

The final review question

What hidden assumption could make this design appear correct while it is wrong? State what the evidence proves, what it does not prove and how a future LILITH can recover from today's mistake.